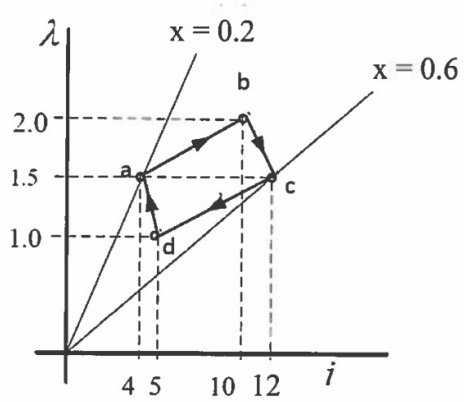
ECE 330

Chapter 5

Analysis of Electromechanical Systems



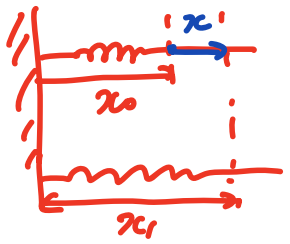
Olga Mironenko
University of Illinois at Urbana-Champaign
Fall 2026

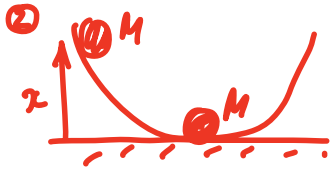
Learning Objectives: you will be able to

- Explain the concepts of **Speed Voltage** and **Transformer Voltage** and calculate them for different electromechanical systems
- Understand the fundamentals of **rotational systems**
- Define and identify **electrically linear systems**
- Calculate **forces of electric origin** via **energy method**
- Define **co-energy**, and calculate **forces of electric origin** via **co-energy method**
- Analyze **energy conversion between two points**
- Define and calculate **EFE** for a system moving between two points
- Define and calculate **EFM** for a system moving between two points
- Define an **energy conversion cycle**, and calculate EFE and EFM over a cycle
- Define a **multiple inputs (ports) system**, and calculate energy, co-energy, and forces/torques for multiport systems

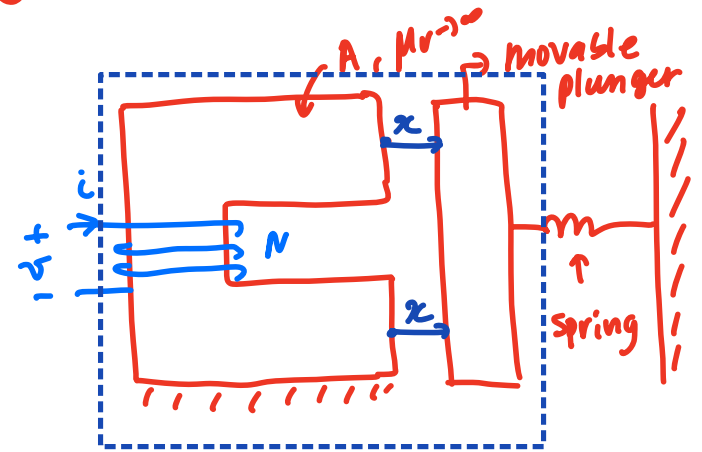
Electromechanical energy conversion: How to make things move using electricity?

* We need to store energy to make things move.

① 

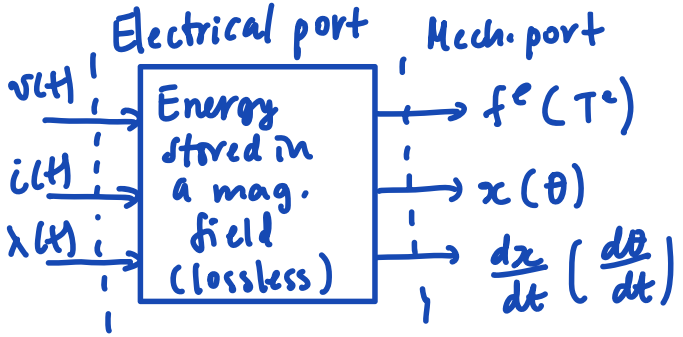
② 

③ Inductors store energy in a mag. field. Where the energy stored? Energy stored in a gap?? energy stored in a core



↑ closed system (lossless, energy is conserved)

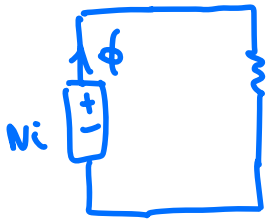
④ Let's abstract general system;



Transformer Voltage and Speed Voltage

4 from electr. port and find $v(t)$:

MEC:



$\mathcal{R}_g(x(t)) \Rightarrow \phi = \dfrac{Ni}{\mathcal{R}_g(x(t))} \Rightarrow \lambda = N\phi = \dfrac{N^2 i}{\mathcal{R}_g(x(t))} \Rightarrow \boxed{\lambda(x,i) = L(x)\cdot i}$

$\mathcal{R}_g(x(t)) = \dfrac{2x(t)}{\mu_0 A}$

$L(x(t))$

Electrically linear

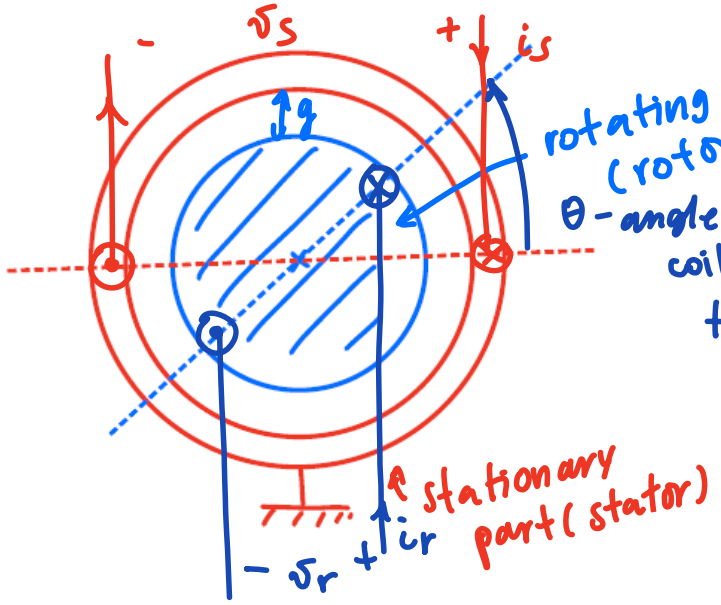
Q: $v(t) - ?$ $\quad v = \dfrac{d\lambda(i,x)}{dt} = \underbrace{\dfrac{\partial\lambda}{\partial i}\dfrac{di}{dt}}_{\text{transformer voltage}} + \underbrace{\dfrac{\partial\lambda}{\partial x}\dfrac{dx}{dt}}_{\text{speed voltage}}$

* Note: If $\lambda(x,i) = L(x)\cdot i$

$$v(t) = L\frac{di}{dt} + i\frac{\partial L}{\partial x}\frac{dx}{dt}$$

* Note: if $x = \text{const}$, $v = L\dfrac{di}{dt}$

Rotational Systems



- $\lambda_s = L_s i_s + \underbrace{(M\cos\theta)}_{L_m(\theta)} i_r$
- $\lambda_r = M\cos\theta\, i_s + L_r i_r$

Electrically Linear System: Relating i and λ

Example 1

Example 2

Problem 1 (25 points) (Spring 2019, M2, p1(a))

A single-phase generator consists of a coil on the stator and a coil on the rotor with a mutual inductance variation as $0.1\cos(\theta)$ (Henries) where θ is the angle from the stator field axis to the rotor field axis . The rotor is being driven at a constant speed of 377 radians per second and the rotor coil has a constant dc current $i_r = 5$ A. The self inductances of the stator and rotor are both constants and you may assume a linear magnetic core.

a) Compute the open-circuit stator voltage ($i_s = 0$) as a function of time (recall that $d\theta/dt=\omega$ and assume some angle $\theta= \theta_o$ at time zero)

Energy stored in coupling field

Mathematical formulation

What is the force of electric origin?

Calculating the energy

Key Assumption

Evaluating the Integrals

Example 3

- Let $\lambda = L(x)i$. What is the energy stored in the coupling field?

Example 4

Conservation of Power

Define the Co-Energy

Substitute Back into Conservation of Power

Calculating the Co-energy

Evaluating the Integrals

Example 5

- Let $\lambda = L(x)i$. What is the co-energy stored in the coupling field?

Example 6

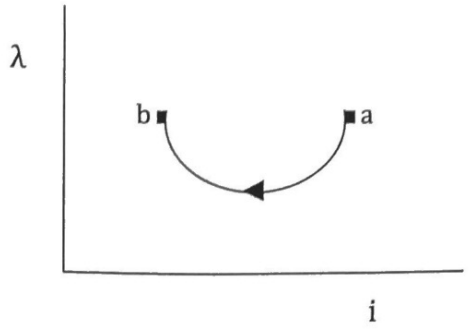
Energy vs. Co-energy

Energy conversion between two points

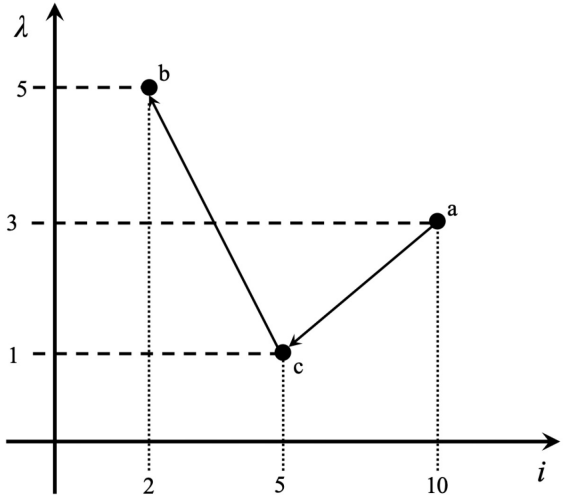
Example 7

Example 7-cont

Example 8



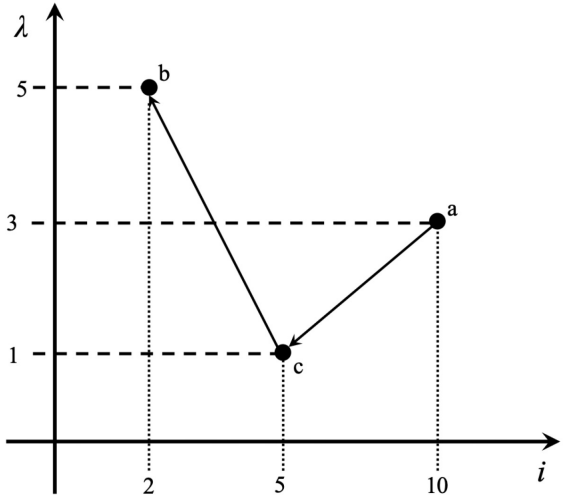
Example 9



An electromechanical system with $\lambda = L(x)i$ is moved from point a to b through point c as shown. The current is given as Amps and the flux linkage is given as Amp turns.

Energy over a cycle

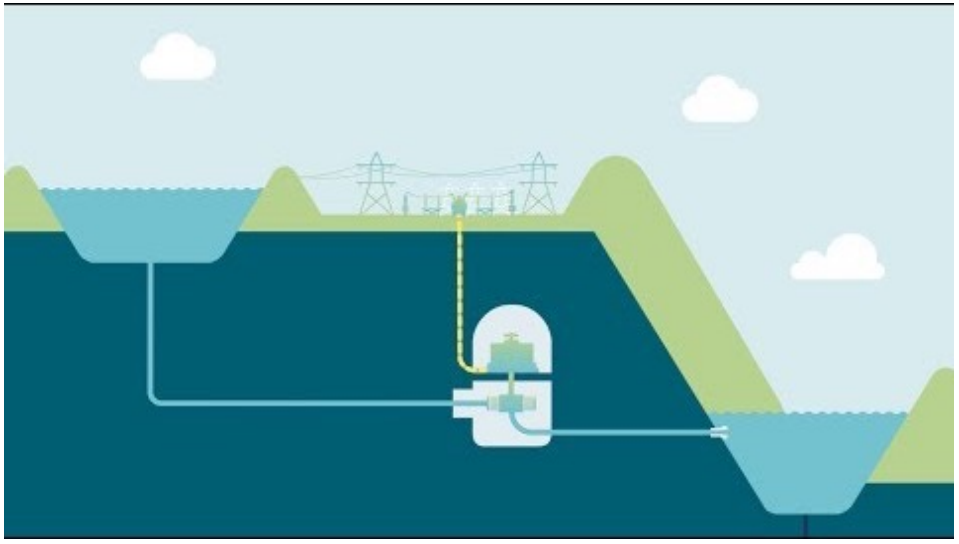
Example 9-cont



An electromechanical system with $\lambda = L(x)i$ is moved from point a to b through point c as shown.
The current is given as Amps and the flux linkage is given as Amp turns.

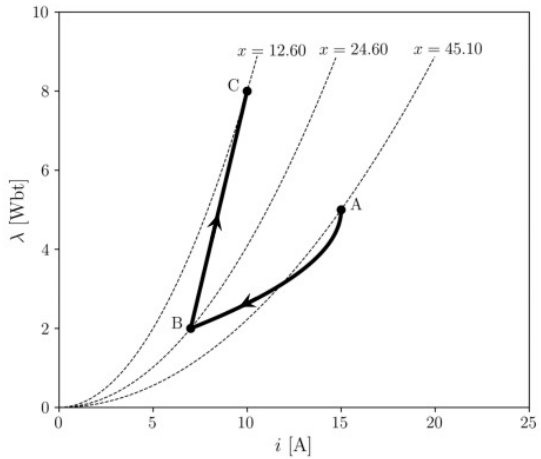
Flipping Direction

- Flipping the cycle direction converts:
  - Motor into generator
  - Generator into motor
  - Example: pumped hydro storage



https://www.youtube.com/watch?v=_PH0IJ-_qOI

Example 10



An electro-mechanical system that has the flux linkage relationship $\lambda = \frac{1}{x-0.1} i^2$ moves from point A ($i_a = 15$ A, $\lambda_a = 5$ Wbt) to point C ($i_c = 10$ A and $\lambda_c = 8$ Wbt) through the intermediate point B ($i_b = 7$ A, $\lambda_c = 2$ Wbt). The equation for the curve between points A and B is given as

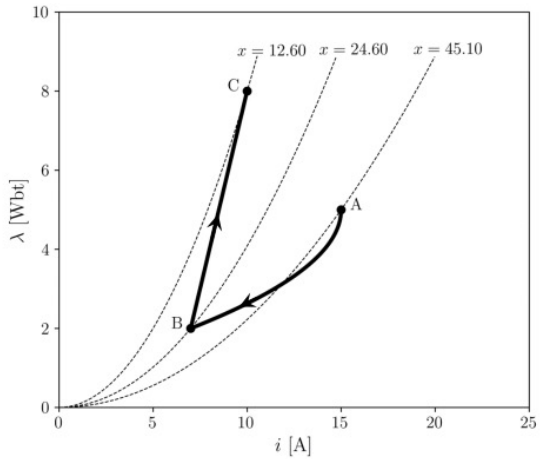
$$i = -\frac{8}{9}(\lambda - 5)^2 + 15$$

and the curve between points B and C is a straight line. The values for x at each given point are also given.

For the movement between these points, find:

a) The energy at point A and C.
b) The EFE when moving from point A to C.
c) The EFM when moving from point A to C.
d) If we move back from point C to point A along a straight line, what is the EFM over this cycle? Is the system behaving like a motor or a generator? Explain.

Example 10-cont



An electro-mechanical system that has the flux linkage relationship $\lambda = \frac{1}{x-0.1} i^2$ moves from point A ($i_a = 15$ A, $\lambda_a = 5$ Wbt) to point C ($i_c = 10$ A and $\lambda_c = 8$ Wbt) through the intermediate point B ($i_b = 7$ A, $\lambda_c = 2$ Wbt). The equation for the curve between points A and B is given as

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Example 11

A certain system has flux linkage given as

$$\lambda = \frac{0.3}{x - 0.01} i^2 \qquad \text{with the constraint that } x > 0.01.$$

Find the energy from the mechanical system (EFM) and the energy from the electrical system (EFE) as the system moves from $x = 0.020$ m to $x = 0.015$ m while i is held constant at $i = 2$ A.

Example 11-cont

Multiple inputs system

Multiple inputs system

Example 12

Example 13